

CHAPTER 2

ENERGY METHODS

2.1 Strain Energy

- Consider a rod BC of length L and uniform cross-sectional area A , which is attached at B to a fixed support, and subjected at C to a slowly increasing axial load P in Figure below.

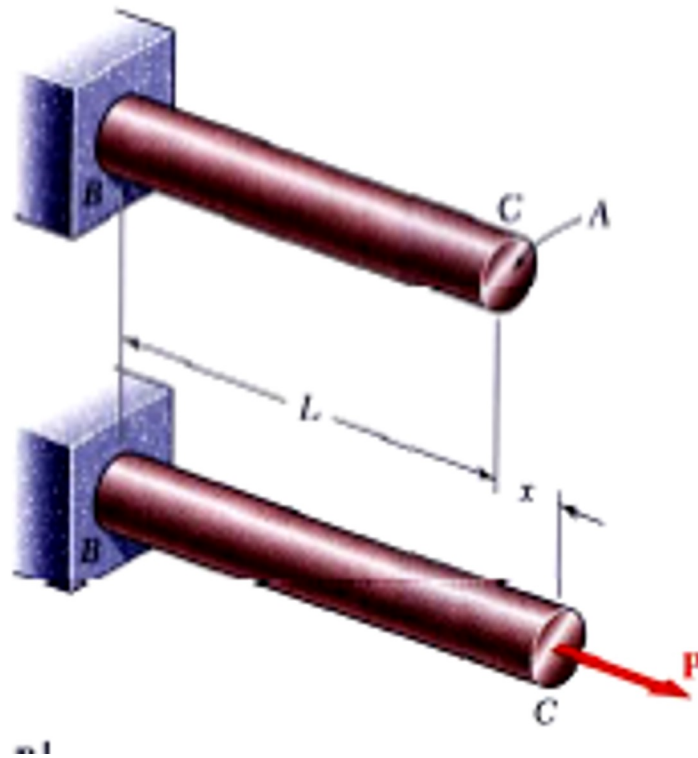


Figure 1

- By plotting the magnitude P of the load against the deformation x of the rod, we obtain a certain load deformation diagram below, that is characteristic of the rod BC.
- Let us consider the work dU done by the load P as the rod elongates by a small amount dx . This elementary work is equal to the product of the magnitude P of the load and of the small elongation dx .
- We write $dU = Pdx$

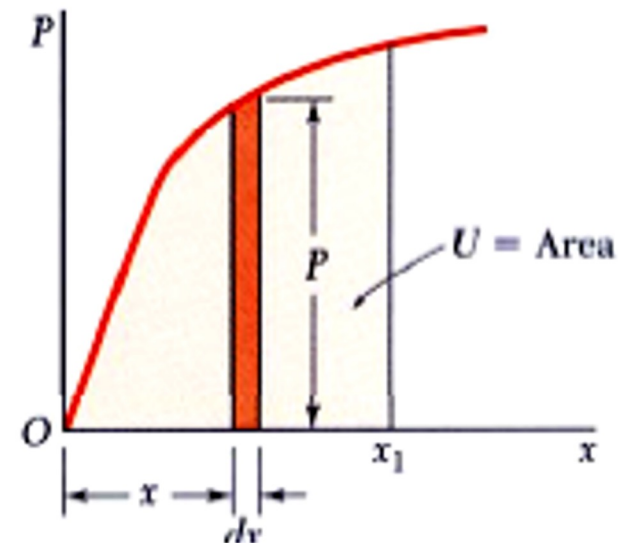


Figure 2

- The expression obtained is equal to the element of area of width $d\mathbf{x}$ located under the load-deformation diagram (Figure.2). The total work U done by the load as the rod undergoes a deformation $\mathbf{x}_1$ is thus

$$U = \int_0^{\mathbf{x}_1} \mathbf{P} d\mathbf{x}$$

Is equal to the area under the load-deformation diagram between $x=0$ and $x=x_1$.

- The work done by the load P as it is slowly applied to the rod must result in the increase of some energy associated with the deformation of the rod. This energy is referred to as the **strain energy** of the rod.

$$\text{Strain Energy} = U = \int_0^{\mathbf{x}_1} \mathbf{P} d\mathbf{x}$$

- In the case of a linear and elastic deformation, the portion of the load-deformation diagram involved can be represented by a straight line of equation $P=kx$.

$$U = \int_0^{x_1} kx \, dx = \frac{1}{2} kx_1^2$$
$$= \frac{1}{2} P_1 x_1$$

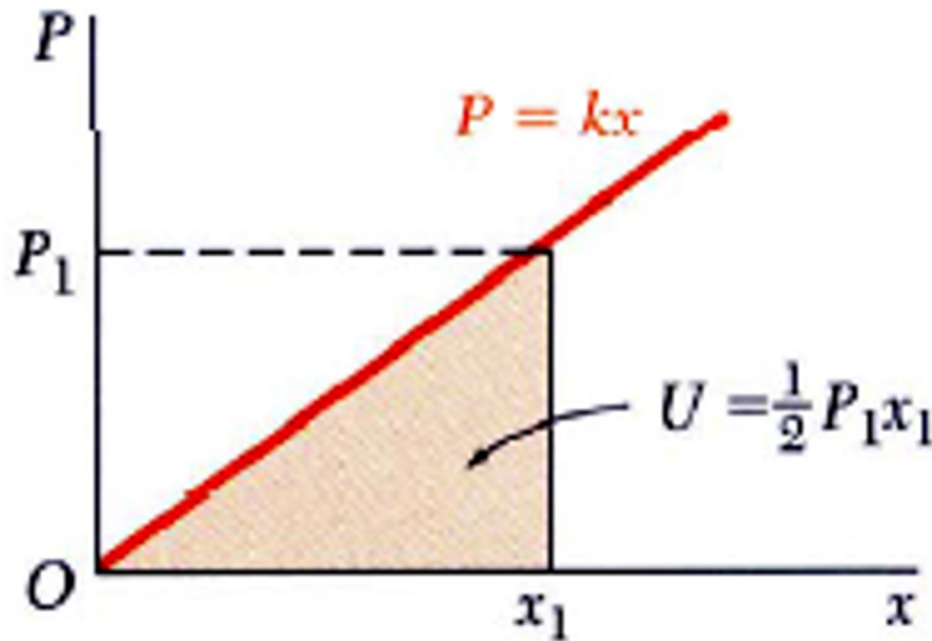


Figure 3

2.3 Strain Energy Density

- In order to eliminate the effect of size, evaluate the strain energy per unit volume will be considered. The strain Energy U by the volume $V=AL$ of the rod.

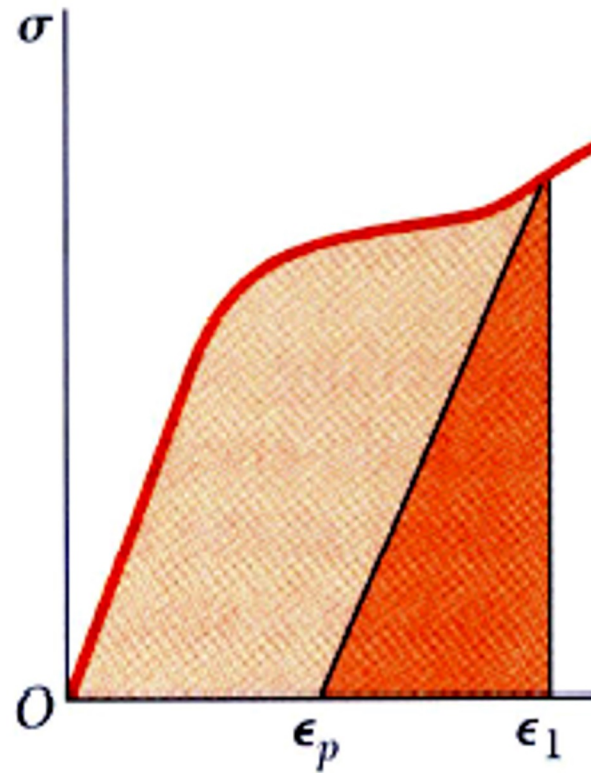
$$\frac{U}{V} = \int_0^{x_1} \frac{P}{A} \frac{dx}{L}$$

- Recalling that P/A represents the normal stress σ_x in the rod, and x/L the normal strain ϵ_x , we write

$$\frac{U}{V} = \int_0^{\epsilon_1} \sigma_x d\epsilon_x$$

- The strain energy per unit volume, U/V , is referred to as the ***Strain-energy density*** and will be denoted by the letter ***u***.

Figure 4



$$\text{Strain Energy} = u = \int_0^{\epsilon_1} \sigma \, d\epsilon$$

- The total strain energy density resulting from the deformation is equal to the area under the curve to ϵ_1 .
- As the material is unloaded, the stress returns zero but there is a permanent deformation. Only the strain energy represented by the triangular area is recovered.
- Remainder of the energy spent in deforming the material is dissipated as heat.

- The strain Energy density resulting from setting $\epsilon_1 = \epsilon_R$ is the *modulus of toughness*.
- The energy per unit volume required to cause the material to rupture is related to its ductility as well as its ultimate strength.

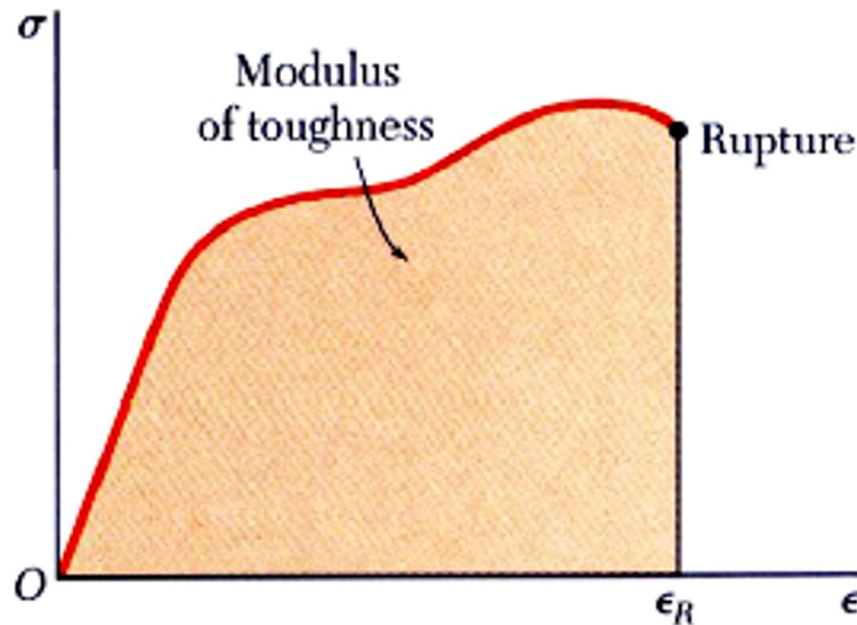


Figure 5

- If the stress remains within the proportional limit,

$$u = \int_0^{\epsilon_1} E \epsilon_x d\epsilon_x = \frac{E \epsilon_1^2}{2} = \frac{\epsilon_1^2}{2E}$$

- The strain Energy density resulting from setting $\sigma_1 = \sigma_Y$ is the modulus of resilience.

$$u_Y = \frac{\epsilon_Y^2}{2E} = \text{modulus of resilience}$$

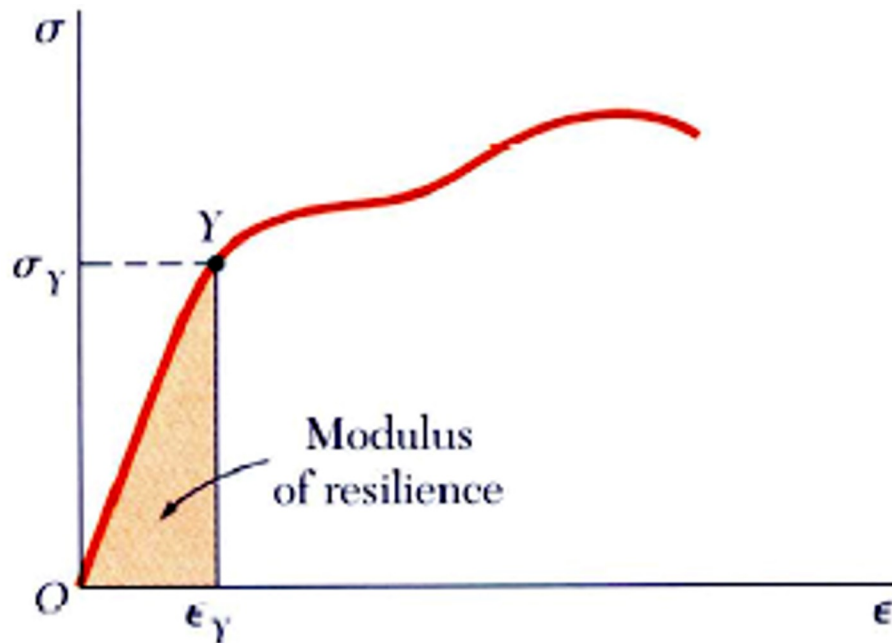


Figure 6

2.4 Elastic strain Energy for Normal stresses

In an element with a nonuniform stress distribution

$$u = \lim_{\Delta V \rightarrow 0} \frac{\Delta U}{\Delta V} = \frac{dU}{dV}$$

$$u = \int_0^{\epsilon_x} \sigma_x d\epsilon_x$$

For values of $u < u_Y$, i.e below the proportional limit,

$$u = \frac{1}{2} E \epsilon_x^2 = \frac{1}{2} \sigma_x \epsilon_x$$

$$U = \int \frac{1}{2} \frac{\sigma_x^2}{E} dV = \text{Elastic strain Energy}$$

Strain Energy under Axial Loading

- Under axial loading, $\sigma_x = P/A$. $dV = Adx$

$$U = \int_0^L \frac{P^2}{2AE} dx$$

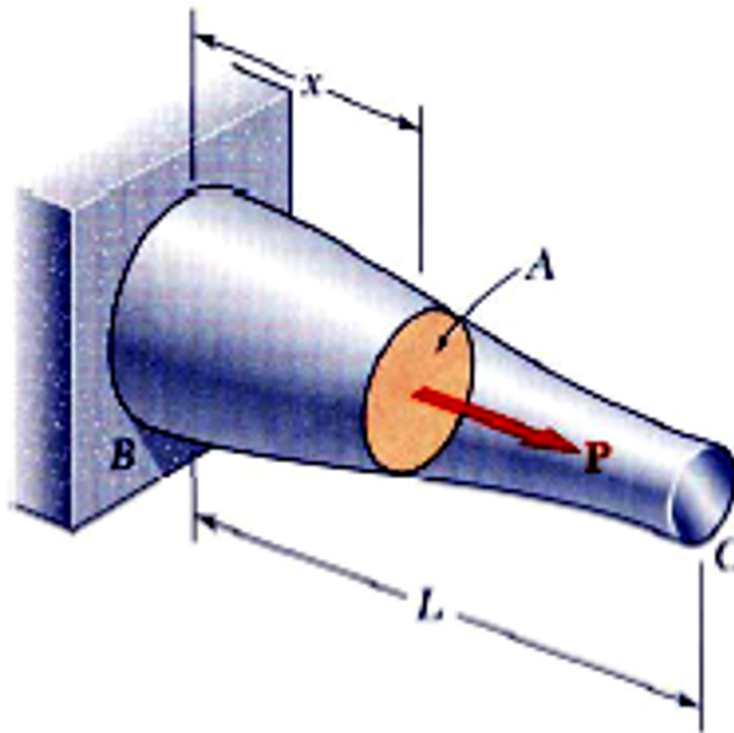


Figure 7

- For a rod of uniform cross section,

$$U = \frac{P^2 L}{2AE}$$

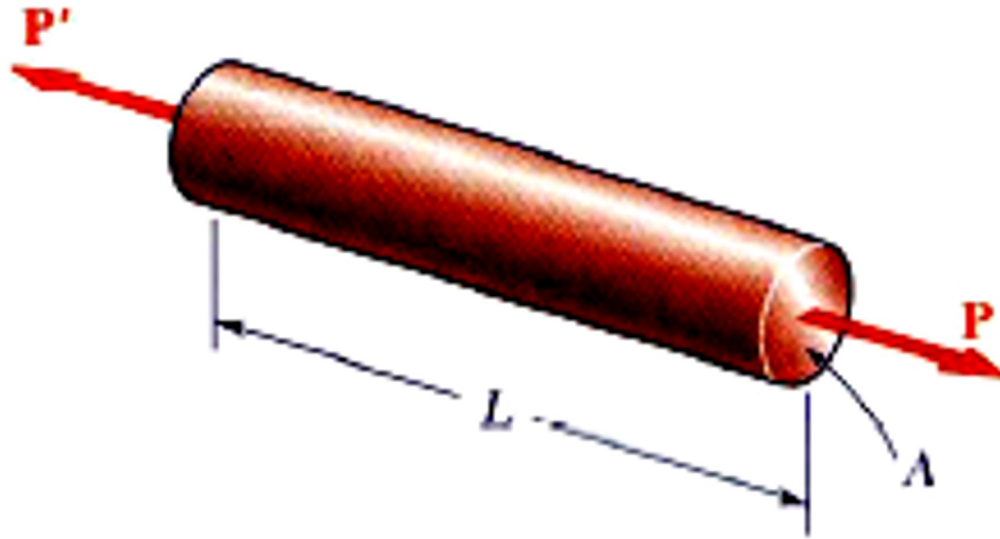


Figure 8.

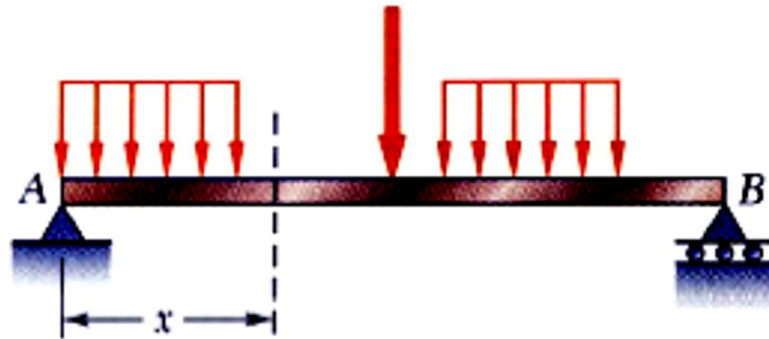
Strain Energy in Bending

- For a beam subjected to a bending load,

$$U = \int \frac{\sigma_x^2}{2E} dV = \int \frac{M^2 y^2}{2EI^2} dV$$

Setting $dV = dA dx$,

$$U = \int_0^L \int_A \frac{M^2 y^2}{2EI^2} dA dx = \int_0^L \frac{M^2}{2EI^2} \left[\int_A y^2 dA \right] dx$$
$$= \int_0^L \frac{M^2}{2EI} dx$$



$$\sigma_x = \frac{M y}{I}$$

- For an end-loaded cantilever beam,

$$M = -Px$$

$$U = \int_0^L \frac{P^2 x^2}{2EI} dx = \frac{P^2 L^3}{6EI}$$



Figure 10.

2.5 Strain Energy for shearing Stresses

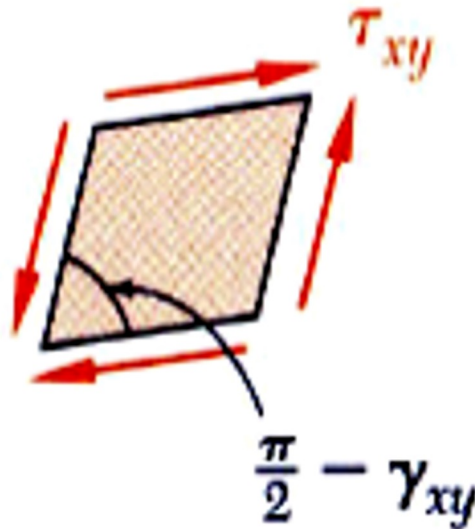
- For material subjected to plane shearing stresses,

$$u = \int_0^{\gamma_{xy}} \tau_{xy} d\gamma_{xy}$$

- For values of τ_{xy} within the proportional limit,

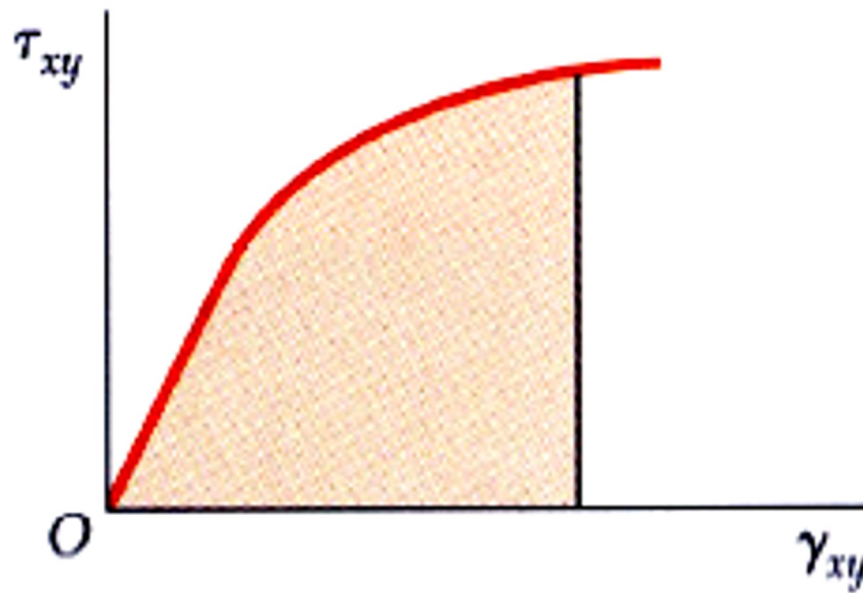
$$u = \frac{1}{2} G \gamma_{xy}^2 = \frac{1}{2} \tau_{xy} \gamma_{xy} = \frac{\tau_{xy}^2}{2G}$$

- G is the modulus of rigidity of the material



- The total strain Energy is found from

$$U = \int u dV = \int \frac{\tau_{xy}^2}{2G} dV$$



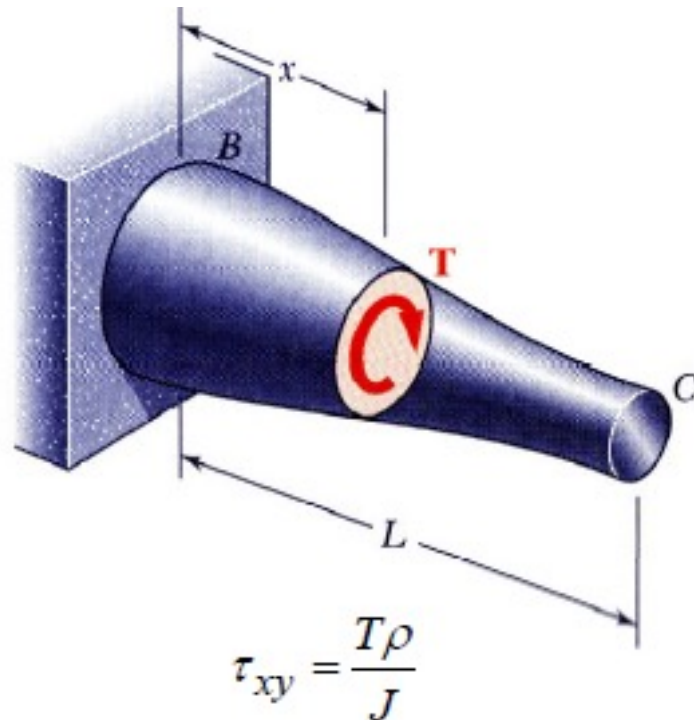
Strain Energy in Torsion

- For a shaft subjected to torsion load,

$$U = \int \frac{\tau_{xy}^2}{2G} dV = \int \frac{T^2 \tau^2}{2GJ^2} dV$$

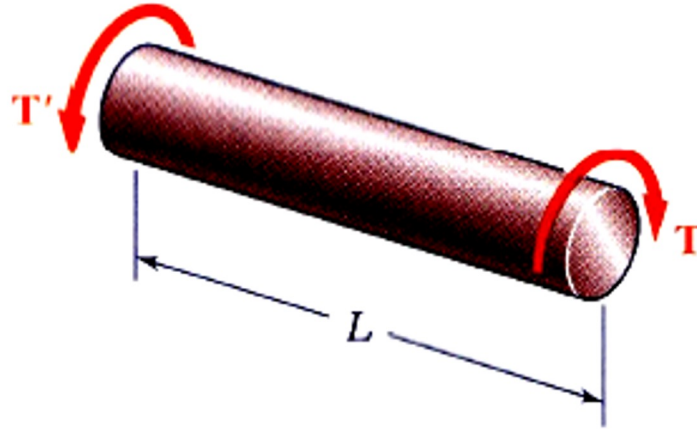
- Setting $dV = dA \, dx$,

$$U = \int_0^L \int_A \frac{T^2 \tau^2}{2GJ^2} dA \, dx = \int_0^L \frac{T^2}{2GJ^2} \left(\int_A \tau^2 dA \right) dx = \int_0^L \frac{T^2}{2GJ} dx$$



- In the case of a uniform shaft,

$$U = \frac{T^2 L}{2GJ}$$



2.6 Strain Energy for a General state of stress

- Previously we found strain energy due to uni-axial stress and plane shearing stress. For a general state of stress,

$$u = \frac{1}{2} (\sigma_x \epsilon_x + \sigma_y \epsilon_y + \sigma_z \epsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{zx} \gamma_{zx})$$

With respect to the principal axes for an elastic, isotropic body,

$$u = \frac{1}{2E} [\sigma_a^2 + \sigma_b^2 + \sigma_c^2 - 2\nu(\sigma_a \sigma_b + \sigma_b \sigma_c + \sigma_c \sigma_a)]$$

$$= u_v + u_d$$

$$u_v = \frac{1-2\nu}{6E} (\sigma_a + \sigma_b + \sigma_c)^2 = \text{due to volume change}$$

$$u_d = \frac{1}{12G} [(\sigma_a - \sigma_b)^2 + (\sigma_b - \sigma_c)^2 + (\sigma_c - \sigma_a)^2] = \text{due to distortion}$$

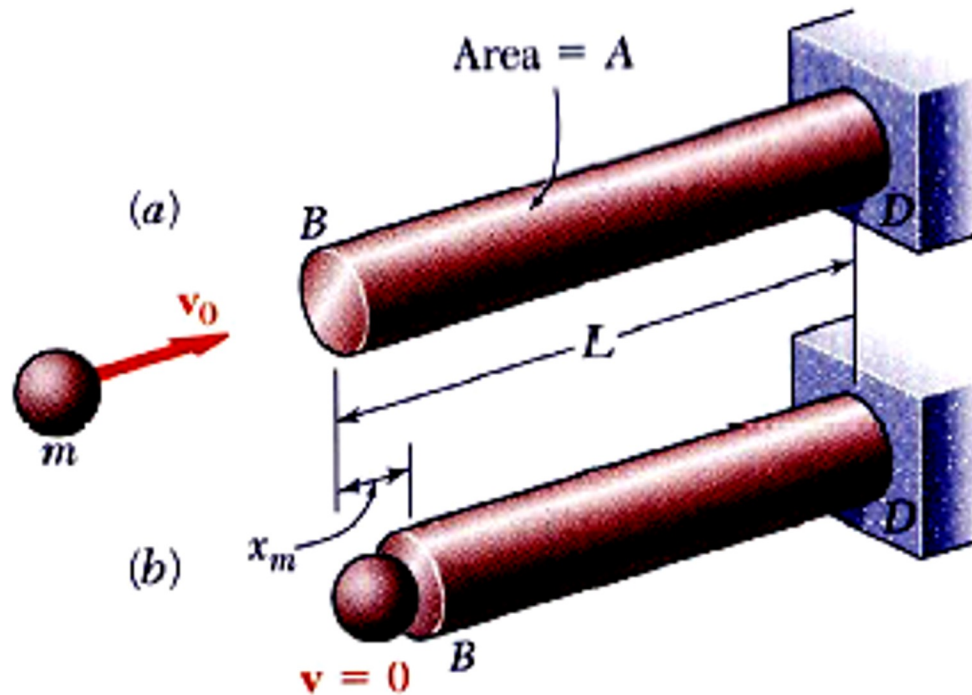
- Basis for the maximum distortion energy failure criteria,

$$u_d < (u_d)_Y = \frac{\sigma_Y^2}{6G} \text{ for a tensile test specimen}$$

2.7 Work done by IMPACT LOADING

- Consider a rod which is hit at its end with a body of mass m moving with a velocity v_0 .
- Rod deforms under impact. Stresses reach a maximum value σ_m and then disappear.
- To determine the maximum stress σ_m
 1. Assume that the kinetic energy is transferred entirely to the structure,
$$U_m = \frac{1}{2}mv_o^2$$
 2. Assume that the stress-strain diagram obtained from a static test is also valid under impact loading.

- Figure 11



- Maximum value of the strain energy,

$$U_m = \int \frac{\sigma_m^2}{2E} dV$$

- For the case of a uniform rod,

$$\sigma_m = \sqrt{\frac{2U_m E}{V}} = \sqrt{\frac{mv_o^2 E}{V}}$$

Design for Impact Loads

Maximum stress can be reduced by:

- Uniformity of stress
- Low modulus of elasticity with high yield strength
- High volume

- For the case of a uniform rod,

$$\square_m = \sqrt{\frac{2U_m E}{V}}$$

- For the case of the non-uniform rod,

$$\square_m = \sqrt{\frac{16U_m E}{5AL}}$$

$$V = 4A(L/2) + A(L/2) = 5AL/2$$

$$\square_m = \sqrt{\frac{8U_m E}{V}}$$

- For the case of the cantilever beam

$$\square_m = \sqrt{\frac{6U_m E}{L(I/c^2)}}$$

$$L(I/c^2) = L\left(\frac{1}{4} \square c^2 L\right) = \frac{1}{4} V$$

$$\square_m = \sqrt{\frac{24U_m E}{V}}$$

2.8 Work and Energy under a single Load

- Previously, we found the strain energy by integrating the energy density over the volume. For a uniform rod,

$$\begin{aligned} U &= \int u dV = \int \frac{\epsilon^2}{2E} dV \\ &= \int_0^L \frac{(P_1 / A)^2}{2E} A dx = \frac{P_1^2 L}{2AE} \end{aligned}$$

- Strain energy may also be found from the work of the single load P_1 .

$$U = \int_0^{x_1} P dx$$

- For an elastic deformation,

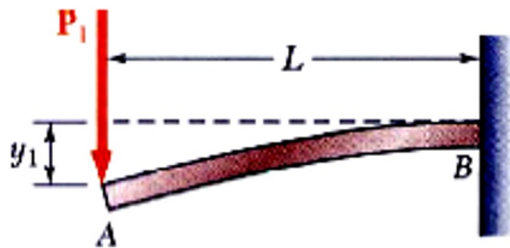
$$U = \int_0^{x_1} P dx = \int_0^{x_1} kx dx = \frac{1}{2} kx_1^2 = \frac{1}{2} P_1 x_1$$

- Knowing the relationship between force and displacement,

$$x_1 = \frac{P_1 L}{AE}$$

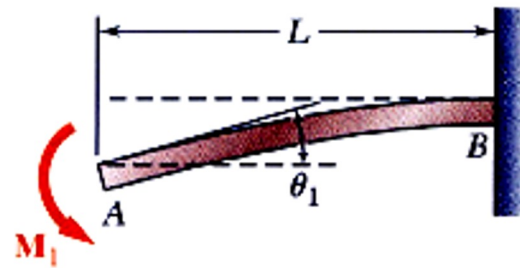
$$U = \frac{1}{2} P_1 \left(\frac{P_1 L}{AE} \right) = \frac{P_1^2 L}{2AE}$$

- Strain energy may be found from the work of other types of single concentrated loads.



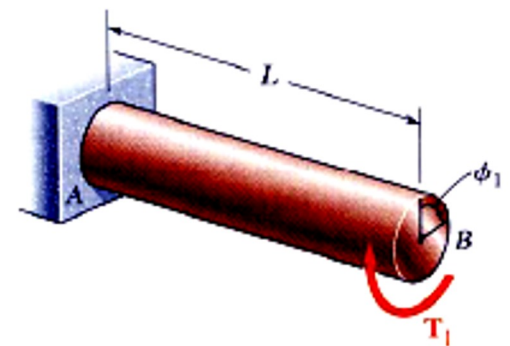
$$U = \int_0^{y_1} P dy = \frac{1}{2} P_1 y_1$$

$$= \frac{1}{2} P_1 \left(\frac{P_1 L^3}{3EI} \right) = \frac{P_1^2 L^3}{6EI}$$



$$U = \int_0^{\theta_1} M d\theta = \frac{1}{2} M_1 \theta_1$$

$$= \frac{1}{2} M_1 \left(\frac{M_1 L}{EI} \right) = \frac{M_1^2 L}{2EI}$$

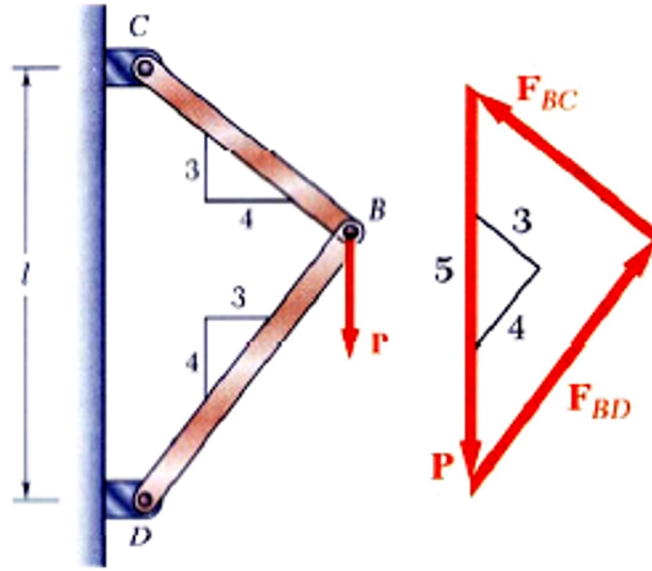


$$U = \int_0^{\phi_1} T d\phi = \frac{1}{2} T_1 \phi_1$$

$$= \frac{1}{2} T_1 \left(\frac{T_1 L}{JG} \right) = \frac{T_1^2 L}{2JG}$$

Deflection under a single load

- From the given geometry below, $L_{BC}=0.6l$ $L_{BD}=0.8l$



- From statics,

$$F_{BC}=+0.6P \quad F_{BD}=-0.8P$$

- If the strain energy of a structure due to a single concentrated load is known, then the equality between the work of the load and energy may be used to find the deflection.

- Strain energy of the structure,

$$\begin{aligned} U &= \frac{F_{BC}^2 L_{BC}}{2AE} + \frac{F_{BD}^2 L_{BD}}{2AE} \\ &= \frac{P^2 l [(0.6)^3 + (0.8)^3]}{2AE} = 0.364 \frac{P^2 l}{AE} \end{aligned}$$

- Equating work and strain energy,

$$U = 0.364 \frac{P^2 L}{AE} = \frac{1}{2} P y_B$$

$$y_B = 0.728 \frac{Pl}{AE}$$

Work and Energy under several Loads

- Deflections of an elastic beam subjected to two concentrated loads,

$$x_1 = x_{11} + x_{12} = \square_{11} P_1 + \square_{12} P_2$$

$$x_2 = x_{21} + x_{22} = \square_{21} P_1 + \square_{22} P_2$$

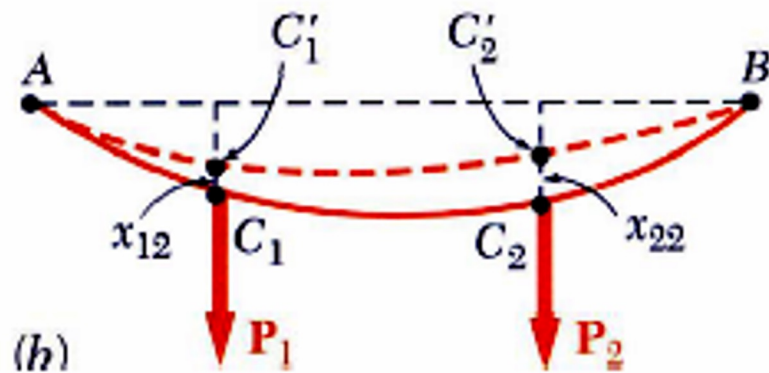
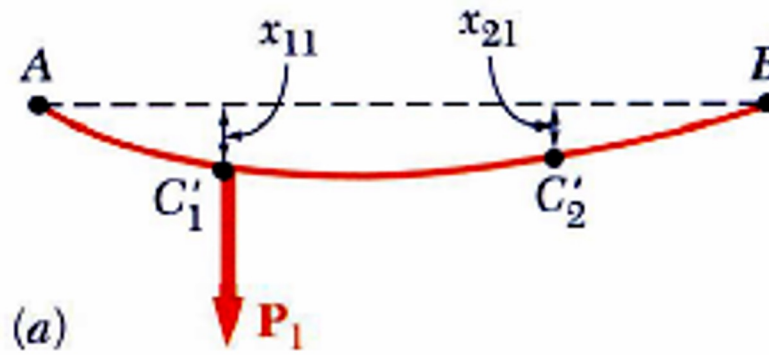
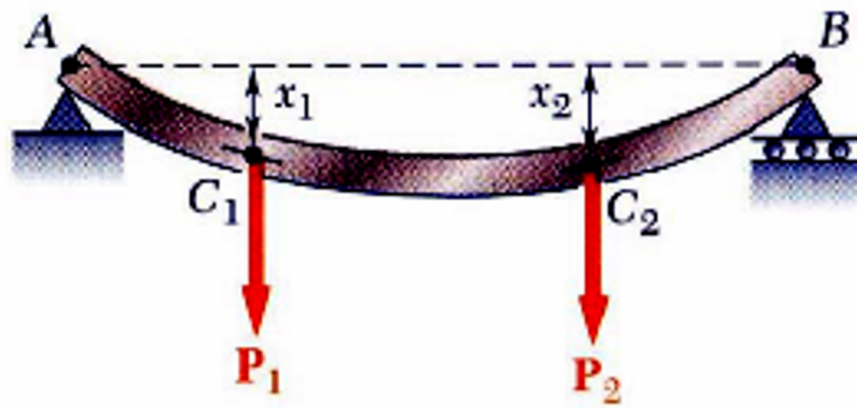
- Compute the strain energy in the beam by evaluating the work done by slowly applying P_1 followed by P_2 ,

$$U = \frac{1}{2} (\square_{11} P_1^2 + 2 \square_{12} P_1 P_2 + \square_{22} P_2^2)$$

- Reversing the application sequence yields,

$$U = \frac{1}{2} (\square_{22} P_2^2 + 2 \square_{21} P_2 P_1 + \square_{11} P_1^2)$$

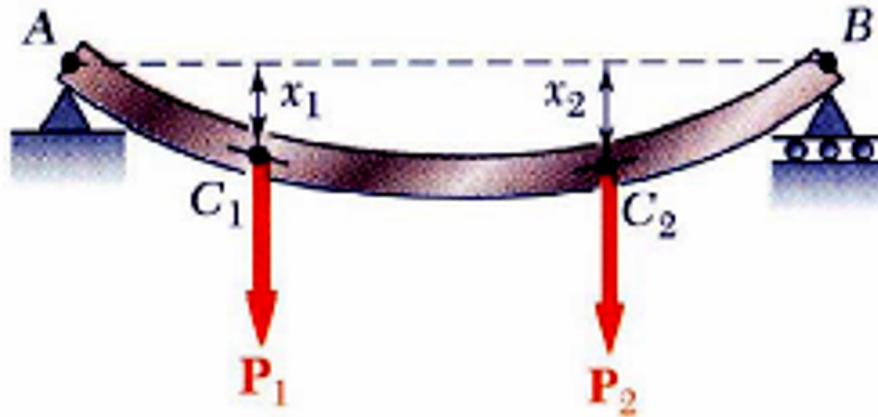
- Strain energy expressions must be equivalent. It follows that $\square_{12} = \square_{21}$ (*Maxwells's reciprocal theorem*)



Castigliano's Theorem

- Strain energy for any elastic structure subjected to two concentrated loads,

$$U = \frac{1}{2} \left(\square_{11} P_1^2 + 2 \square_{12} P_1 P_2 + \square_{22} P_2^2 \right)$$



- Differentiating with respect to the loads,

$$\frac{\partial U}{\partial P_1} = \square_{11} P_1 + \square_{12} P_2 = x_1$$

$$\frac{\partial U}{\partial P_2} = \square_{12} P_1 + \square_{22} P_2 = x_2$$

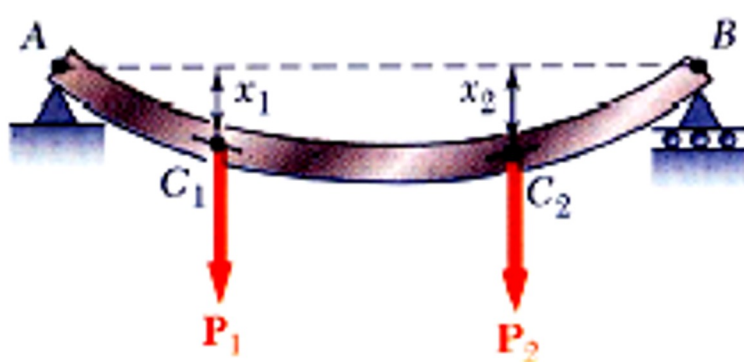
- **Castigliano's theorem:** For an elastic structure subjected to n loads, the deflection x_j of the point of application of P_j can be expressed as

$$x_j = \frac{\partial U}{\partial P_j} \text{ and } \theta_j = \frac{\partial U}{\partial M_j} \quad \square_j = \frac{\partial U}{\partial T_j}$$

Deflections by Castigliano's Theorem

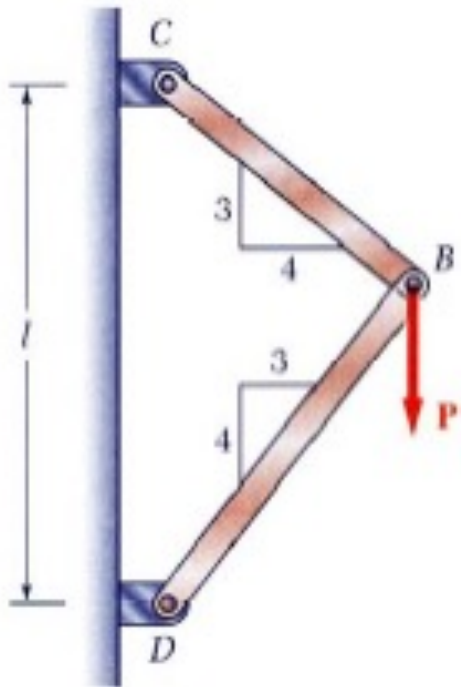
- Application of Castigliano's theorem is simplified if the differentiation with respect to the load P_j is performed before the integration or summation to obtain the strain energy U .

- In the case of a beam,



$$U = \int_0^L \frac{M^2}{2EI} dx \quad x_j = \frac{\partial U}{\partial P_j} = \int_0^L \frac{M}{EI} \frac{\partial M}{\partial P_j} dx$$

- For a truss,



$$U = \sum_{i=1}^n \frac{F_i^2 L_i}{2A_i E} \quad x_j = \frac{\partial U}{\partial P_j} = \sum_{i=1}^n \frac{F_i L_i}{A_i E} \frac{\partial F_i}{\partial P_j}$$

The End